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Valve module for a vacuum pumping system

Abstract

A valve module for a vacuum pumping system, comprising: a plurality of inlets for receiving a fluid; a plurality of pressure sensors, each configured to measure a fluid pressure associated with a respective inlet; a first fluid line manifold; a second fluid line manifold; a plurality of multifurcating conduits, each connecting a respective inlet to both the first and second fluid line manifolds; a plurality of valves disposed in the multifurcating conduits; and a valve controller coupled to the sensors and the valves; wherein the valve controller is configured to control, based on pressure measurements from the sensors, the valves such that a fluid flow through a multifurcating conduit is directed to either only the first fluid line manifold or only the second fluid line manifold.

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Background/Summary

CROSS-REFERENCE OF RELATED APPLICATION
(1) This application is a Section 371 National Stage Application of International Application No. PCT/GB2022/051078, filed Apr. 28, 2022, and published as WO 2022/229642A1 on Nov. 3, 2022, the content of which is hereby incorporated by reference in its entirety and which claims priority of British Application No. 2106098.3, filed Apr. 29, 2021.
FIELD
(2) The present invention relates to valve modules for use with vacuum pumping systems, including but not limited to vacuum systems for pumping fluids from semiconductor processing tool.
BACKGROUND
(3) Semiconductor fabrication plants fabricate integrated circuit chips. In the fabrication of such devices, wafers are processed through a number of different processing stations, including stations at which the wafer undergoes, for example, chemical vapor deposition, physical vapor deposition, implant, etch and lithography processes. Many of these processes involve the use of a gaseous ambient and often require the use of high vacuum and reduced gas pressures.
(4) Vacuum pumps are used to provide these reduced gas pressures in process chambers, provide chamber evacuation, and maintain flows of processing gases.
(5) The discussion above is merely provided for general background information and is not intended to be used as an aid in determining the scope of the claimed subject matter. The claimed subject matter is not limited to implementations that solve any or all disadvantages noted in the background.
SUMMARY
(6) When the pressure inside a chamber of a semiconductor processing tool is not at working vacuum, for example after a gas chamber has been vented to atmospheric pressure to enable service or maintenance, a so-called “pump-down event” is performed to establish the required reduced gas pressure in the chamber. A pump-down event involves pumping gas from the chamber so as to reduce the pressure therein to the required level.
(7) Vacuum and abatement systems may be used to pump gas from multiple gas chambers of a semiconductor processing tool simultaneously using a common pump via a common manifold. The present inventors have realised that in such systems, because multiple chambers are fluidly connected to a common manifold, performing a pump-down event for one of those chambers may affect the conditions within others of those chambers. For example, a pump-down event performed on one chamber may cause highly undesirable fluctuations in other chambers connected to the same manifold.
(8) Aspects of the present invention provide a valve module for controlling fluid from multiple chambers of a semiconductor processing tool in such a way that these deficiencies are reduced or eliminated.
(9) In a first aspect, there is provided a valve module for a vacuum pumping system. The valve module comprises: a plurality of inlets, each inlet of the plurality of inlets being configured to

receive a pumped fluid; a plurality of pressure sensors, each pressure sensor of the plurality of pressure sensors being configured to measure a pressure of a fluid associated with a respective one of the plurality of inlets; a first fluid line manifold; a second fluid line manifold; a plurality of multifurcating conduits, wherein each multifurcating conduit fluidly connects a respective inlet to both of the first fluid line manifold and the second fluid line manifold; a plurality of valves, where a respective one or more valves of the plurality of valves is disposed in a respective one of the plurality of multifurcating conduits; and a valve controller operatively coupled to the plurality of pressure sensors and the plurality of valves; wherein the controller is configured to control, based on pressure measurements received from the plurality of pressure sensors, the plurality of valves such that each of the one or more valves disposed in a respective multifurcating conduit selectably directs a fluid flow through that multifurcating conduit to either only the first fluid line manifold or only the second fluid line manifold.
(10) Each of the multifurcating conduits may comprise a first branch and a second branch, the first branch being fluidly connected to the first fluid line manifold and the second branch being fluidly connected to the second fluid line manifold. Each of the one or more valves disposed in a respective multifurcating conduit may comprise a first valve disposed in the first branch of that multifurcating conduit, and a second valve disposed in the second branch of that multifurcating conduit.
(11) The plurality of pressure sensors may comprise a first pressure sensor configured to measure the pressure of a fluid associated with a first inlet of the plurality of inlets, the first inlet being an inlet of a first multifurcating conduit of the plurality of multifurcating conduits. The valve controller may be configured to, responsive to pressure measurements received from the first pressure sensor fulfilling one or more first criteria, control the one or more valves disposed in the first multifurcating conduit to direct fluid flow through the first multifurcating conduit to the second fluid line manifold. The one or more first criteria may consist of one or more criteria selected from the group of criteria consisting of: the measured pressure exceeding a first threshold value; the measured pressure exceeding the first threshold value for at least a first time period; a rate of increase of the measured pressure exceeding a second threshold value; and a rate of increase of the measured pressure exceeding the second threshold value for at least a second time period. The valve controller may be configured to, responsive to pressure measurements received from the first pressure sensor fulfilling one or more second criteria, controlling the one or more valves disposed in the first multifurcating conduit to direct fluid flow through the first multifurcating conduit to the first fluid line manifold. The one or more second criteria may consist of one or more criteria selected from the group of criteria consisting of: the measured pressure being less than or equal to a third threshold value; the measured pressure being less than the third threshold value for at least a third time period; a rate of decrease of the measured pressure exceeding a fourth threshold value; a rate of decrease of the measured pressure exceeding the fourth threshold value for at least a fourth time period; a predefined time period elapsing.
(12) At least the plurality of inlets, the first fluid line manifold, the second fluid line manifold, the plurality of multifurcating conduits, the plurality of valves, and the valve controller may be configured as a single integrated unit and housed in a frame.
(13) The valve module may further comprise a plurality of further valves, wherein, for each valve of the plurality of valves, a respective pair of further valves is disposed either side of that valve. The further valves may be manually operated valves.
(14) The valve module may further comprise a gas inlet for receiving a gas for purging one or more of the multifurcating conduits and/or actuating one or more of the valves.
(15) In a further aspect, there is provided a system comprising: a semiconductor processing tool comprising a plurality of processing chambers; the valve module of any preceding aspect, wherein each inlet of the plurality of inlets is fluidly coupled to a respective processing chamber of the plurality of processing chambers; and one or more vacuum pumps operating coupled to the first

fluid line manifold and the second fluid line manifold.
(16) The system may further comprise cooling apparatus for providing a cooling fluid to one or more of the processing chambers, wherein the valve module is disposed on top of the cooling apparatus.
(17) In a further aspect, there is provided a method for valve module for a vacuum pumping system, the valve module comprising: receiving, at each inlet of a plurality of inlets, a respective pumped fluid, each inlet being an inlet to a respective multifurcating conduit of a plurality of multifurcating conduits, each multifurcating conduit fluidly connecting a respective inlet to both of a first fluid line manifold and a second fluid line manifold; measuring, by one or more pressure sensors of a plurality of pressure sensors, a pressure of a respective one of the pumped fluids; and controlling, by a controller, based the one or more measured pressures, one or more valves of a plurality of valves, the one or more valves being disposed in a first multifurcating conduit of the plurality of multifurcating conduits; wherein the controlling the one or more valves selectably directs a fluid flow through the first multifurcating conduit to either only the first fluid line manifold or only the second fluid line manifold.
(18) The method may further comprise measuring, by a first pressure sensor of a plurality of pressure sensors, a pressure of a pumped fluid in the first multifurcating conduit, and, responsive to pressure measurements received from the first pressure sensor fulfilling one or more first criteria, controlling the one or more valves disposed in the first multifurcating conduit to direct fluid flow through the first multifurcating conduit to the second fluid line manifold and preventing fluid flow through the first multifurcating conduit to the first fluid line manifold. The method may further comprise, thereafter, responsive to pressure measurements received from the first pressure sensor fulfilling one or more second criteria, controlling the one or more valves disposed in the first multifurcating conduit to prevent fluid flow through the first multifurcating conduit to the second fluid line manifold, and subsequently direct fluid flow through the first multifurcating conduit to the first fluid line manifold.
(19) The Summary is provided to introduce a selection of concepts in a simplified form that are further described in the Detailed Description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic illustration (not to scale) of a semiconductor fabrication facility;
(2) FIG. **2** is a schematic illustration (not to scale) showing a perspective view of a valve module of the semiconductor fabrication facility;
(3) FIG. **3** is a process flow chart showing certain steps of a process of pumping gas in the semiconductor fabrication facility; and
(4) FIG. **4** is a schematic illustration (not to scale) showing a system in which two valve modules are mounted on top of a plurality of cooling apparatuses.
DETAILED DESCRIPTION
(5) FIG. **1** is a schematic illustration (not to scale) of a semiconductor fabrication facility **100**, in accordance with an embodiment.
(6) The semiconductor fabrication facility **100** comprises a semiconductor processing tool **102**, a valve module **104**, and a plurality of vacuum pumps **106**.
(7) The semiconductor processing tool **102** comprises a plurality of process chambers **108** in which semiconductor wafers undergo respective processes. Examples of such processes include, but are not limited to, chemical vapor deposition, physical vapor deposition, implant, etch and lithography

processes.
(8) The plurality of vacuum pumps **106** are configured to pump fluids (i.e. process gases) out of the process chambers **108** of the semiconductor processing tool **102** via the valve module **104**.
(9) The valve module **104** comprises a plurality of inlets **110**, a plurality of multifurcating conduits **112**, a first fluid line manifold **114**, and a second fluid line manifold **116**.
(10) Each of the inlets **110** is fluidly connected to a respective process chamber **108**, such that a pumped fluid may be received from that process chamber **108**.
(11) Each multifurcating conduit **112** fluidly connects a respective inlet **110** to both of the first fluid line manifold **114** and the second fluid line manifold **116**. More specially, in this embodiment, the multifurcating conduits **112** are bifurcating conduits comprising respective first and second branches **118**, **120**. The first branch **118** of each multifurcating conduit **112** fluidly connects the respective inlet **110** to the first fluid line manifold **114**. The second branch **120** of each multifurcating conduit **112** fluidly connects the respective inlet **110** to the second fluid line manifold **116**.
(12) The valve module **104** further comprises a plurality of pressure sensors **122**. Each pressure sensor **122** is operatively coupled to a respective inlet **110**, or to a respective multifurcating conduit **112** at or proximate to an inlet **110**.
(13) Each pressure sensor **122** is configured to measure a pressure associated with a respective process chamber **108**. In particular, each pressure sensor **122** is configured to measure a pressure of a process gas that is being pumped out of a respective process chamber **108**. It is preferable that the pressure sensors **122** are located as close as possible to the outlets of the process chambers **108**.
(14) The valve module **104** further comprises a plurality of gate valves, and more specifically a plurality of first gate valves **124** and a plurality of second gate valves **126**. In this embodiment, the first gate valves **124** and the second gate valves **126** are pneumatic valves.
(15) Each of the first gate valves **124** is disposed on a respective one of the first branches **118**, and is configured to control the flow of fluid therethrough.
(16) Each of the second gate valves **126** is disposed on a respective one of the second branches **120**, and is configured to control the flow of fluid therethrough.
(17) The valve module **104** further comprises a valve controller **128**.
(18) The valve controller **128** is operatively coupled, via wired or wireless connections (not shown), to each of the plurality of pressure sensors **122** such that pressure measurements taken by the plurality of pressure sensors **122** may be received by the valve controller **128**.
(19) The valve controller **128** is further operatively coupled, via respective pneumatic lines (not shown), to each of the first gate valves **124** and each of the of second gate valves **126**.
(20) As described in more detail later below with reference to FIG. **3**, the valve controller **128** is configured to control operation of the first and second gate valves **124**, **126**, based on the pressure measurements received from the pressure sensors **122**. The valve controller **128** is configured to control operation of the first and second gate valves **124**, **126** by transferring pneumatic fluid thereto via the pneumatic lines.
(21) The valve module **104** further comprises a plurality of manual valves (i.e. valve that are configured to be operated manually by a human operator), and more specifically a plurality of first manual valves **130**, a plurality of second manual valves **132**, and a plurality of third manual valves **134**.
(22) In this embodiment, each first manual valve **130** is disposed on a respective multifurcating conduit **112** between the pressure sensor **122** of that multifurcating conduit **112** and the point at which that multifurcating conduit **112** bifurcates.
(23) In this embodiment, each second manual valve **132** is disposed on a respective first branch **118** of a multifurcating conduit **112** between the first gate valve **124** of that multifurcating conduit **112** and the first fluid line manifold **114**.
(24) In this embodiment, each third manual valve **134** is disposed on a respective second branch

120 of a multifurcating conduit **112** between the second gate valve **126** of that multifurcating conduit **112** and the second fluid line manifold **116**.
(25) Thus, in this embodiment, each of the first and second gate valves **124**, **126** is disposed between a respective pair of manual valves **130-134**. In particular, each first gate valve **124** is disposed between a first manual valve **130** and a second manual valve **132**. Also, each second gate valve **126** is disposed between a first manual valve **130** and a third manual valve **134**.
(26) In this embodiment, the first fluid line manifold **114** is a manifold via which process gases are pumped from process chambers **108** in which semiconductor fabrication processes are being conducted. The first fluid line manifold **114** may be considered to be a “process gas line”. The second fluid line manifold **116** may be considered to be a “pump-down gas line”. The fluid line manifolds **114** and **116** are suitably sized for the gas flow and vacuum requirements.
(27) A pump-down event may be performed to evacuate gas from one or more of the process chambers **108**, which may be at atmospheric pressure, to reduce the pressure therein to a level suitable for a semiconductor fabrication process. The gases evacuated from a gas chamber during pump-down are, for convenience, hereinafter referred to as pump-down gases. In this embodiment, the second fluid line manifold **116** is a manifold via which pump-down gases are pumped out of the process chambers **108**.
(28) Apparatus, including the valve controller **128**, for implementing the above arrangement, and performing the method steps to be described below, may be provided by configuring or adapting any suitable apparatus, for example one or more computers or other processing apparatus or processors, and/or providing additional modules. The apparatus may comprise a computer, a network of computers, or one or more processors, for implementing instructions and using data, including instructions and data in the form of a computer program or plurality of computer programs stored in or on a machine readable storage medium such as computer memory, a computer disk, ROM, PROM etc., or any combination of these or other storage media.
(29) FIG. **2** is a schematic illustration, not to scale, showing a perspective view of the valve module **104**.
(30) In this embodiment, certain components of the valve module **104**, including, for example, at least the inlets **110**, the multifurcating conduits **112**, the first fluid line manifold **114**, the second fluid line manifold **116**, the gate valves **124**, **126**, the valve controller **128**, and the manual valves **130**, **132**, **134** are configured or arranged as a single, integrated unit, hereinafter referred to a the “first integrated unit”. These components are housed in a common frame **200**. The frame **200** may be made of steel.
(31) In some embodiments, the pressure sensors **122** are also comprised in the first integrated unit, and may be housed in the frame **200**. However, in some embodiments, the pressure sensors **122** are separate to the first integrated unit. For example, the pressure sensors **122** may be configured or arranged as a separate, second integrated unit that may be coupled to the first integrated unit, e.g. to the top of the first integrated unit. The second integrated unit comprising the pressure sensors **122** may be coupled between the process chambers **108** and the inlets **110** of the first integrated unit.
(32) FIG. **3** is a process flow chart showing certain steps of a process **300** of pumping gas in the semiconductor fabrication facility **100**.
(33) It should be noted that certain of the process steps depicted in the flowchart of FIG. **3** and described below may be omitted or such process steps may be performed in differing order to that presented below and shown in FIG. **3**. Furthermore, although all the process steps have, for convenience and ease of understanding, been depicted as discrete temporally-sequential steps, nevertheless some of the process steps may in fact be performed simultaneously or at least overlapping to some extent temporally.
(34) At step s**302**, semiconductor fabrication processes are performed in the process chambers **108**. These semiconductor fabrication processes generate process gases.
(35) In this embodiment, at this stage, the first gate valves **124** are open and the second gate valves

126 are closed. Also, all of the manual valves **130-134** are open.
(36) At step s**304**, the vacuum pump **106** coupled to the first fluid line manifold **114** pumps the generated process gases out of the process chambers **108** via the valve module **104**. In particular, in this embodiment, processes gases are pumped from each process chamber **108** and through, in turn, the inlet **110** coupled thereto, the first branch **118** of the multifurcating conduits **112** coupled thereto (including through the first gate valve **124** disposed thereon), and the first fluid line manifold **114**.
(37) At step s**306**, the pressure sensors **122** measure pressures associated with the process chambers **108**. In particular, each pressure sensor **122** measures a pressure of a process gas that is being pumped through a respective inlet **110**. In this embodiment, the pressure sensors **122** measure the pressures substantially continuously.
(38) At step s**308**, the pressure sensors **122** send the measured pressure values to the valve controller **128**. The valve controller **128** processes the received measured pressure values substantially continuously.
(39) At step s**310**, one of the process chambers **108** (hereinafter referred to as “the first process chamber **108**” for convenience) is shut down for inspection, servicing, repair, or maintenance. In this embodiment, the shutting down the first process chamber **108** comprises stopping pumping gas from the first process chamber **108**. In this embodiment, this may be achieved by the operator closing an isolating valve in inlet **110** associated with the first process chamber **108**. In this embodiment, the shutting down the first process chamber **108** further comprises increasing the pressure in the first process chamber **108** to approximately atmospheric pressure. This may be achieved by opening a valve coupled to the first process chamber **108**, thereby allowing air to enter into the first process chamber **108**.
(40) At step s**312**, a human operator performs an inspection, servicing, repair, or maintenance operation on the first process chamber **108**.
(41) Following the inspection, servicing, repair, or maintenance operation, a low gas pressure environment is to be re-established in the first process chamber **108** such that semiconductor fabrication processes may be performed therein.
(42) Accordingly, at step s**314**, the isolating valve associated with the first process chamber **108** is reopened, thereby allowing gases to be pumped from the first process chamber **108**.
(43) This pumping of gases from the first process chamber **108** at step s**314** is a pump-down event.
(44) At step s**316**, the valve controller **128**, processing the measured pressure values received from the pressure sensors **122**, determines that a pump-down event is occurring.
(45) In particular, in this embodiment, the valve controller **128** determines that a pump-down event is occurring for the first process chamber **108** in response to the measured pressure associated with the first process chamber **108** exceeding a first threshold value and/or a calculated rate of increase of the measured pressure associated with first process chamber **108** exceeding a second threshold value.
(46) The first threshold value may be any appropriate threshold valve. The second threshold value may be any appropriate threshold valve.
(47) In some embodiments, the valve controller **128** determines that the pump-down event is occurring for the first process chamber **108** in response to the measured pressure associated with first process chamber **108** exceeding the first threshold value for at least a first time period. The first time period may be any appropriate time period.
(48) In some embodiments, the valve controller **128** determines that the pump-down event is occurring for the first process chamber **108** in response to the calculated rate of increase of the measured pressure associated with first process chamber **108** exceeding the second threshold value for at least a second time period. The second time period may be any appropriate time period.
(49) At step s**318**, responsive to detecting the pump-down event for the first process chamber **108**, the valve controller **128** controls the first gate valve **124** associated with the first process chamber

108 to close. Thus, gas flow from the first process chamber **108** to the first fluid line manifold **114** is prevented or opposed.

(50) In this embodiment, the valve controller **128** conveys pneumatic fluid (e.g. nitrogen) to the first gate valve **124** thereby to control the first gate valve **124**.

(51) At step s**320**, following closure of the first gate valve **124**, the valve controller **128** controls the second gate valve **126** associated with the first process chamber **108** to open. Thus, gas flow from the first process chamber **108** to the second fluid line manifold **116** is permitted.

(52) In this embodiment, the valve controller **128** conveys pneumatic fluid to the second gate valve **126** thereby to control the second gate valve **126**.

(53) At step s**322**, the vacuum pump **106** coupled to the second fluid line manifold **116** pumps the pump-down gas out of the first process chamber **108** via the valve module **104**. In particular, in this embodiment, pump-down gas is pumped from the first process chamber **108** and through, in turn, the inlet **110** coupled thereto, the second branch **120** of the multifurcating conduits **112** coupled thereto (including through the open second gate valve **126** disposed thereon), and the second fluid line manifold **116**.

(54) Thus, pump-down gas is pumped out of the first process chamber **108** thereby to establish a low gas pressure or vacuum environment therein.

(55) At step s**324**, the valve controller **128**, processing the measured pressure values received from the pressure sensors **122**, determines that the pump-down event has ended.

(56) In particular, in this embodiment, the valve controller **128** determines that a pump-down event for the first process chamber **108** has ended in response to the measured pressure associated with the first process chamber **108** being less than or equal to a third threshold value and/or a calculated rate of decrease of the measured pressure associated with first process chamber **108** being more than or equal to a second fourth value. Alternatively, a pump-down event is ended after it has been running for a predefined time period.

(57) The third threshold value may be any appropriate threshold value. In some embodiments, the third threshold value is equal to, or less than, the first threshold value.

(58) The fourth threshold value may be any appropriate threshold valve. In some embodiments, the fourth threshold value is equal to, or less than, the second threshold value.

(59) In some embodiments, the valve controller **128** determines that the pump-down event for the first process chamber **108** has ended in response to the measured pressure associated with first process chamber **108** being less than or equal to the third threshold value for at least a third time period. The third time period may be any appropriate time period.

(60) In some embodiments, the valve controller **128** determines that the pump-down event for the first process chamber **108** has ended in response to the calculated rate of decrease of the measured pressure associated with first process chamber **108** being more than or equal to the fourth threshold value for at least a fourth time period. The fourth time period may be any appropriate time period.

(61) At step s**326**, responsive to detecting the pump-down event for the first process chamber **108** has ended, the valve controller **128** controls the second gate valve **126** associated with the first process chamber **108** to close. Thus, gas flow from the first process chamber **108** to the second fluid line manifold **116** is prevented or opposed.

(62) At step s**328**, following closure of the second gate valve **126**, the valve controller **128** controls the first gate valve **124** associated with the first process chamber **108** to open. Thus, gas flow from the first process chamber **108** to the first fluid line manifold **114** is permitted.

(63) At step s**330**, semiconductor fabrication processes may be performed in the first process chambers **108**. These semiconductor fabrication processes generate process gases.

(64) At step s**332**, the vacuum pump **106** coupled to the first fluid line manifold **114** pumps the generated process gases out of the first process chamber **108** via the valve module **104**.

(65) Thus, a process **300** of pumping gas in the semiconductor fabrication facility **100** is provided.

(66) The above-described system and method advantageously tends to reduce or eliminate pump-

down events detrimentally affecting the conditions within parallel gas chambers. This tends to be achieved by pumping pump-down gases to a separate manifold that is different to that which process gases are pumped to.
(67) Advantageously, pump-down events, and the ending of pump-down events, tend to be detected and mitigated against automatically.
(68) Advantageously, the above-described valve module may be integrated in-line with horizontal manifolds connecting the semiconductor processing tool to the vacuum pumps.
(69) Advantageously, the above-described valve module tends to be robust. The vacuum module may be fully assembled, leak checked, and pre-tested, for example, off-site prior to delivery to a semiconductor fabrication facility, or on-site when delivered. This tends to simplify the installation process and reduce installation time.
(70) Advantageously, the above-described valve module tends to be modular and scalable.
(71) Advantageously, the components in the gas streams of the valve module tend to be easy to service, repair or replace. For example, each gate valve can be isolated from fluid flow by closing the manual valve upstream and downstream of that gate valve, allowing a human operator to service, repair or replace the gate valve.
(72) Advantageously, the status and operating condition of the system tends to be easily monitorable, for example, either via a Human Machine Interface of the valve module, or remotely.
(73) Advantageously, each valve module in a system tends to be easily controllable by a system controller, for example using a communication protocol such as EtherCAT or ethernet.
(74) Advantageously, the above-described valve module allows for multiple mounting options. For example, the valve module may be suspended from a ceiling of a semiconductor fabrication facility, which provides a benefit of not consuming floor space. Alternatively, the valve module can be mounted in a floor-standing frame or on top of other equipment.
(75) What will now be described is an embodiment in which the valve module is mounted on top of other equipment, specifically cooling apparatus for controlling the temperature of the process chambers of the semiconductor processing tool.
(76) FIG. **4** is a schematic illustration (not to scale) showing a system **400** in which two valve modules **104** are mounted on top of a plurality of cooling apparatuses **402**. The cooling apparatuses **402** are typically referred to as “chiller racks” or “chillers”.
(77) The system **400** comprises six cooling apparatuses **402**, two valve modules **104**, a power source **404**, and a pneumatic source **406**.
(78) In this embodiment, the valve modules **104** may be substantially identical to those described above with reference to FIGS. **1** and **2**. Each valve module **104** is configured to receive a respective plurality of pumped fluid flows from process chambers **108** fluidly coupled thereto.
(79) Each cooling apparatus **402** is fluidly coupled to a respective process chamber **108**. Each cooling apparatus **402** is configured to supply a flow of a cooling fluid to the respective process chamber **108** to which it is coupled. The cooling fluid may be used in the process chambers **108** for controlling temperature.
(80) In this embodiment, the valve modules **104** and the cooling apparatuses **402** are arranged in a stacked configuration. More specifically, each valve module is disposed on top of three of the cooling apparatuses **402**, which themselves are positioned adjacent to one another, e.g. in a side-by-side configuration.
(81) Advantageously, the stacked arrangement provides a reduced footprint within a semiconductor fabrication facility.
(82) Furthermore, the stacked arrangement tends to facilitate coupling of the cooling apparatuses **402** and the valve modules **104** to the process chambers **108**. For example, the stacked arrangement tends to allow for closer positioning of the cooling apparatuses **402** and/or the valve modules **104** with respect to the process chambers **108**, thereby reducing conduit lengths and, consequently, installation time and difficulty. In addition, the likelihood of leakage from or damage to conduits

may be reduced due to their reduced lengths.
(83) In this embodiment, the power source **404** is electrically coupled to each of the cooling apparatuses **402** and the valve modules **104**. The power source **404** is configured to supply electrical power to each of the cooling apparatuses **402** and the valve modules **104**. Accordingly, the power source **404** may be considered to be a common power source.
(84) Advantageously, use of a common power source for the cooling apparatuses **402** and the valve modules **104** facilitates installation and tends to provide for reduced footprint and cabling.
(85) In this embodiment, the pneumatic source **406** is fluidly coupled, via one or more conduits, to each of the cooling apparatuses **402** and the valve modules **104**. The pneumatic source **406** is configured to supply a pneumatic fluid to each of the cooling apparatuses **402** and the valve modules **104**. Accordingly, the pneumatic source **406** may be considered to be a common pneumatic source. The pneumatic fluid may be any appropriate type of gas including, but not limited to, nitrogen gas, or CDA (Clean Dry Air).
(86) Advantageously, use of a common pneumatic source **406** for the cooling apparatuses **402** and the valve modules **104** facilitates installation and tends to provide for reduced footprint and pneumatic fluid conduit length.
(87) In this embodiment, in the valve modules **104**, the pneumatic fluid received from the pneumatic source **406** may be used to actuate valves of the valve modules **104**. More specifically, the valve controller **128** of a valve module **104** may be configured to convey the pneumatic fluid, via respective pneumatic lines, to each of the first gate valves **124** and each of the of second gate valves **126**, thereby to actuate the first and second gate valves **124**, **126**. Thus, the pneumatic fluid may be considered to be a “valve control fluid”.
(88) In this embodiment, in the valve modules **104**, the pneumatic fluid received from the pneumatic source **406** may be used to perform a purge process, to purge a portion of one or more of the multifurcating conduits **112**. More specifically, a manual operator is able to convey the pneumatic fluid into each of the multifurcating conduits **112** via a respective purge port in each of the multifurcating conduits **112**. The pneumatic fluid may be forced through at least a portion of a multifurcating conduit **112** thereby to purge the at least a portion of a multifurcating conduit **112**. The pneumatic fluid may exit a multifurcating conduit **112** via the first fluid line manifold **114** and/or the second fluid line manifold **116**. Thus, the pneumatic fluid may be considered to be a “purge fluid”. Purging may typically be performed prior to maintaining or servicing the valve module **104**, e.g. replacing first gate valve **124** and/or second gate valve **126**. Advantageously, a first gate valve **124** and/or a second gate valve **126** may be isolated from the rest of the system by closing the first manual valve **130**, the second manual valve **132**, and the third manual valve **134**.
(89) In this embodiment, the purge ports in the valve modules **104** may be used to perform a leak test on one or more of the multifurcating conduits **112**. More specifically, a valve module **104** may further comprise means for detecting a leak from the multifurcating conduits **112** using the purge ports, or a human operator may detect the presence of a leak using appropriate sensing equipment attached to the purge ports.
(90) In the embodiment shown in FIG. **4**, there are six cooling apparatuses **402** and two valve modules **104**. However, in other embodiments, the system may comprise a different number of cooling apparatuses and/or a different number of valve modules.
(91) In the embodiment shown in FIG. **4**, each of the valve modules **104** is mounted on top of three cooling apparatuses **402**. However, in other embodiments, one or more of the valve modules may be mounted on top of a different number of cooling apparatuses. In some embodiments, one or more cooling apparatuses are mounted on top of one or more valve modules or other equipment.
(92) In the above embodiments, the valve module is implemented in a semiconductor fabrication facility for routing pumped process gases. However, in other embodiments, the valve module may be implemented in a different system and be used for routing a different type of fluid.
(93) In the above embodiments, there is a single semiconductor processing tool which comprises

six gas chambers. However, in other embodiments, there is more than one semiconductor processing tool. One or more of the semiconductor processing tool may comprise a different number of gas chambers, other than six.
(94) In the above embodiments, there is either a single valve module, or in the embodiment of FIG. **4**, two valve modules. However, in other embodiments, there may be a different number of valve modules.
(95) In the above embodiments, a valve module comprises six inlets and six multifurcating conduits. However, in other embodiments, the valve module comprises a different number of inlets and multifurcating conduits other than six.
(96) In the above embodiments, each multifurcating conduit comprises two gate valves, one on each branch. However, in other embodiments, a multifurcating conduit comprises a different number of gate valves other than two. In some embodiments, a multifurcating conduit comprises a single valve (e.g. a three-way valve) operable to direct fluid flow along a select branch on the multifurcating conduit. In some embodiments, multiple gate valves are arranged along each branch. In some embodiments, a multifurcating conduit comprises more than two branches, each of which may include a respective one or more gate valves.
(97) In the above embodiments, each multifurcating conduit comprises three manual valves. However, in other embodiments, a multifurcating conduit comprises a different number of manual valves other than three. For example, in some embodiments, the manual valve may be omitted. In some embodiments, a multifurcating conduit comprises more than three manual valves arranged along the multifurcating conduit in any appropriate way.
(98) Although elements have been shown or described as separate embodiments above, portions of each embodiment may be combined with all or part of other embodiments described above.
(99) Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are described as example forms of implementing the claims.

Claims

1. A valve module for a vacuum pumping system, the valve module comprising: a plurality of inlets, each inlet of the plurality of inlets being configured to receive a pumped fluid; a plurality of pressure sensors, each pressure sensor of the plurality of pressure sensors being configured to measure a pressure of a fluid associated with a respective one of the plurality of inlets; a first fluid line manifold; a second fluid line manifold; a plurality of multifurcating conduits, wherein each multifurcating conduit fluidly connects a respective inlet to both of the first fluid line manifold and the second fluid line manifold; a plurality of valves, where a respective one or more valves of the plurality of valves is disposed in a respective one of the plurality of multifurcating conduits; and a valve controller operatively coupled to the plurality of pressure sensors and the plurality of valves; wherein the valve controller is configured to control, based on pressure measurements received from the plurality of pressure sensors, the plurality of valves such that each of the one or more valves disposed in a respective multifurcating conduit selectably directs a fluid flow through that multifurcating conduit to either only the first fluid line manifold or only the second fluid line manifold.
2. The valve module of claim 1, wherein: each of the multifurcating conduits comprises a first branch and a second branch, the first branch being fluidly connected to the first fluid line manifold and the second branch being fluidly connected to the second fluid line manifold; and each of the one or more valves disposed in a respective multifurcating conduit comprises a first valve disposed in the first branch of that multifurcating conduit, and a second valve disposed in the second branch of that multifurcating conduit.

3. The valve module of claim 1, wherein: the plurality of pressure sensors comprises a first pressure sensor configured to measure the pressure of a fluid associated with a first inlet of the plurality of inlets, the first inlet being an inlet of a first multifurcating conduit of the plurality of multifurcating conduits; and the valve controller is configured to, responsive to pressure measurements received from the first pressure sensor fulfilling one or more first criteria, control the one or more valves disposed in the first multifurcating conduit to direct fluid flow through the first multifurcating conduit to the second fluid line manifold.
4. The valve module of claim 3, wherein the one or more first criteria consist of one or more criteria selected from the group of criteria consisting of: the measured pressure exceeding a first threshold value; the measured pressure exceeding the first threshold value for at least a first time period; a rate of increase of the measured pressure exceeding a second threshold value; and a rate of increase of the measured pressure exceeding the second threshold value for at least a second time period.
5. The valve module of claim 3, wherein the valve controller is configured to, responsive to pressure measurements received from the first pressure sensor fulfilling one or more second criteria, controlling the one or more valves disposed in the first multifurcating conduit to direct fluid flow through the first multifurcating conduit to the first fluid line manifold.
6. The valve module of claim 5, wherein the one or more second criteria consist of one or more criteria selected from the group of criteria consisting of: the measured pressure being less than or equal to a third threshold value; the measured pressure being less than the third threshold value for at least a third time period; a rate of decrease of the measured pressure exceeding a fourth threshold value; a rate of decrease of the measured pressure exceeding the fourth threshold value for at least a fourth time period; a predefined time period elapsing.
7. The valve module of claim 1, wherein at least the plurality of inlets, the first fluid line manifold, the second fluid line manifold, the plurality of multifurcating conduits, the plurality of valves, and the valve controller are configured as a single integrated unit and housed in a frame.
8. The valve module of claim 1, further comprising a plurality of further valves, wherein, for each valve of the plurality of valves, a respective pair of further valves is disposed either side of that valve.
9. The valve module of claim 8, wherein the further valves are manually operated valves.
10. The valve module of claim 1, further comprising a gas inlet for receiving a gas for purging one or more of the multifurcating conduits and/or actuating one or more of the valves.
11. A system comprising: a semiconductor processing tool comprising a plurality of processing chambers; the valve module of claim 1, wherein each inlet of the plurality of inlets is fluidly coupled to a respective processing chamber of the plurality of processing chambers; and one or more vacuum pumps operating coupled to the first fluid line manifold and the second fluid line manifold.
12. The system of claim 11, further comprising cooling apparatus for providing a cooling fluid to one or more of the processing chambers, wherein the valve module is disposed on top of the cooling apparatus.
13. A method for valve module for a vacuum pumping system, the valve module comprising: receiving, at each inlet of a plurality of inlets, a respective pumped fluid, each inlet being an inlet to a respective multifurcating conduit of a plurality of multifurcating conduits, each multifurcating conduit fluidly connecting a respective inlet to both of a first fluid line manifold and a second fluid line manifold; measuring, by one or more pressure sensors of a plurality of pressure sensors, a pressure of a respective one of the pumped fluids; and controlling, by a controller, based the one or more measured pressures, one or more valves of a plurality of valves, the one or more valves being disposed in a first multifurcating conduit of the plurality of multifurcating conduits; wherein the controlling the one or more valves selectably directs a fluid flow through the first multifurcating conduit to either only the first fluid line manifold or only the second fluid line manifold.

14. The method of claim 13, further comprising: measuring, by a first pressure sensor of a plurality of pressure sensors, a pressure of a pumped fluid in the first multifurcating conduit; and responsive to pressure measurements received from the first pressure sensor fulfilling one or more first criteria, controlling the one or more valves disposed in the first multifurcating conduit to direct fluid flow through the first multifurcating conduit to the second fluid line manifold and preventing fluid flow through the first multifurcating conduit to the first fluid line manifold.

15. The method of claim 14, further comprising thereafter, responsive to pressure measurements received from the first pressure sensor fulfilling one or more second criteria, controlling the one or more valves disposed in the first multifurcating conduit to prevent fluid flow through the first multifurcating conduit to the second fluid line manifold, and subsequently direct fluid flow through the first multifurcating conduit to the first fluid line manifold.